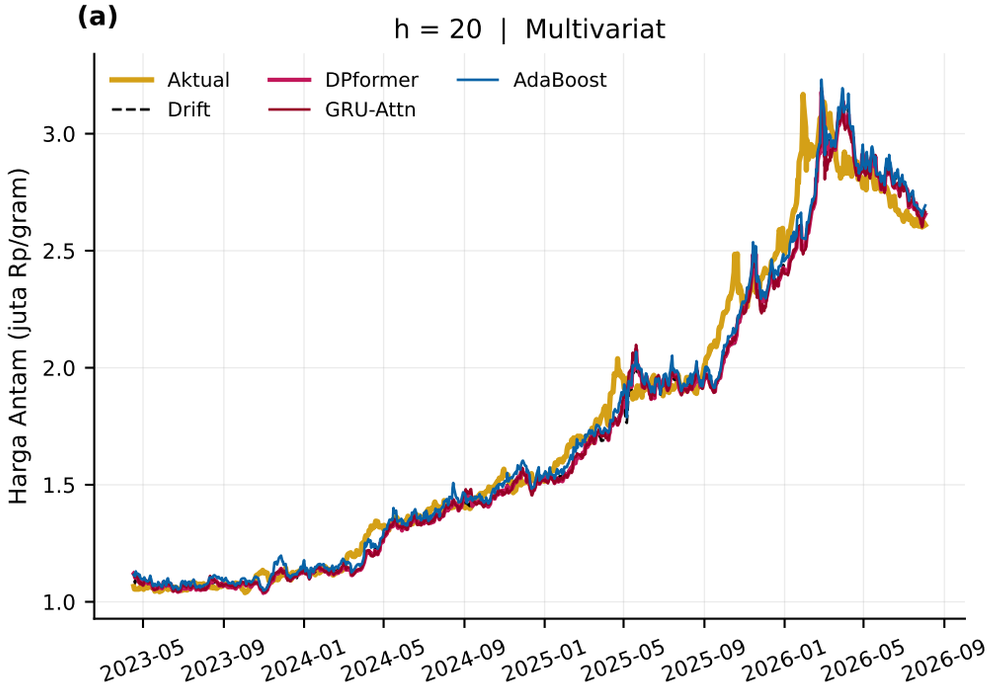
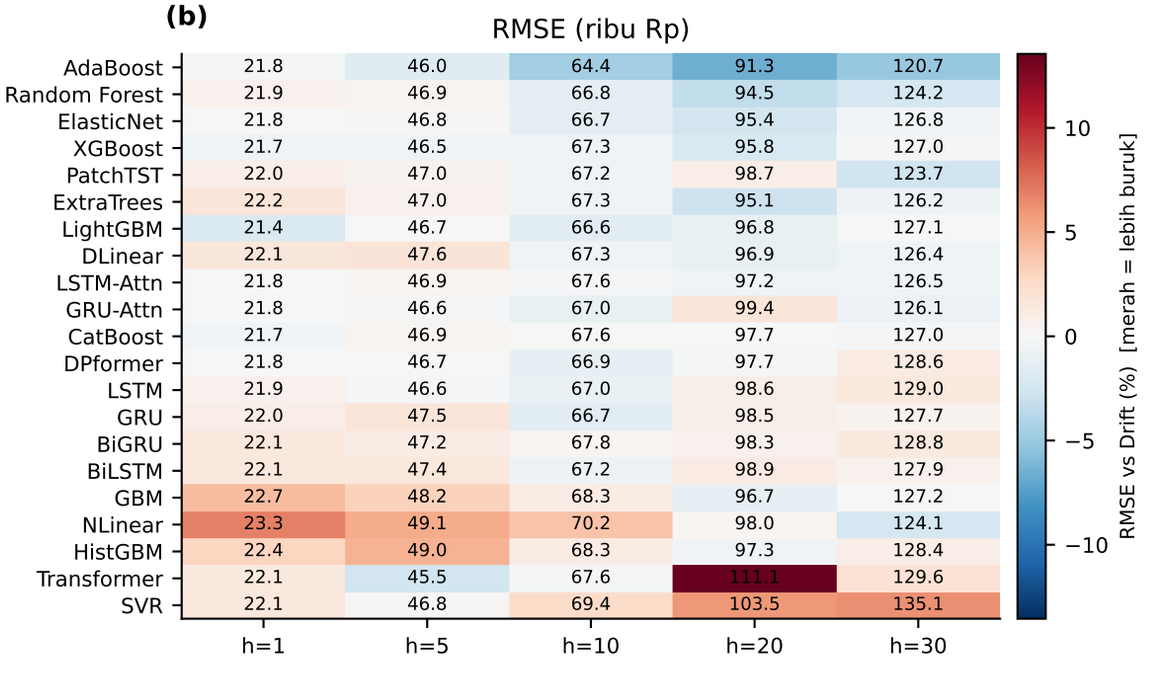
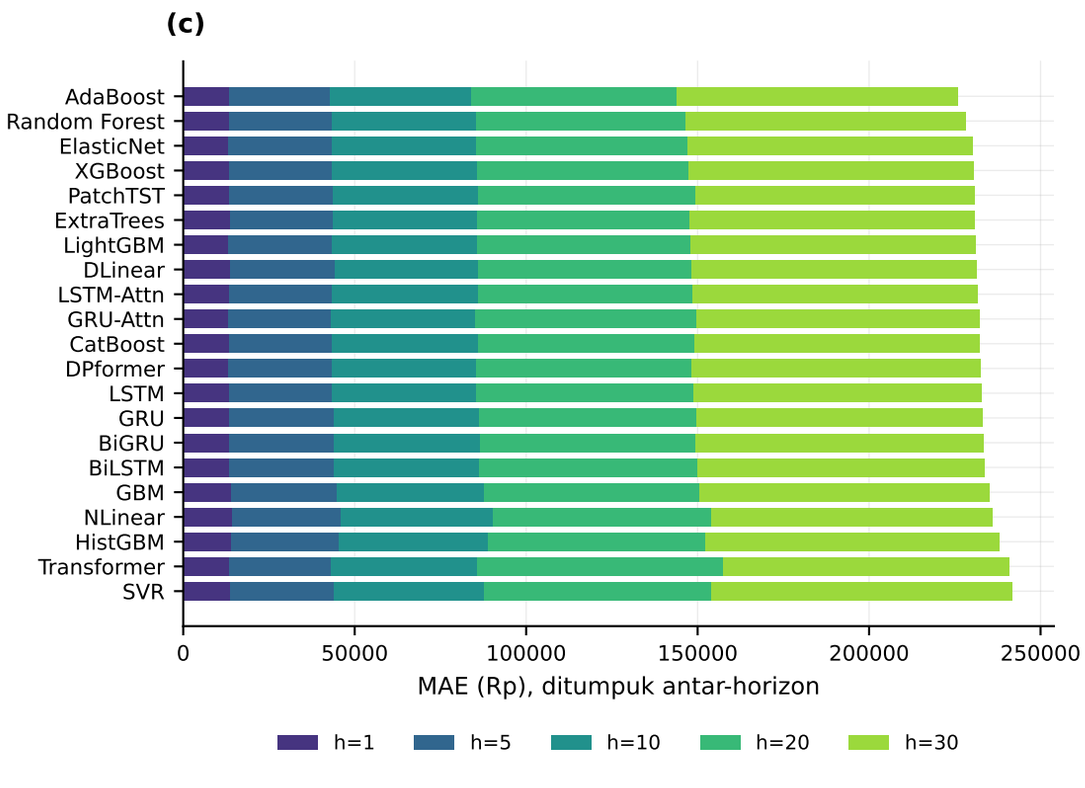
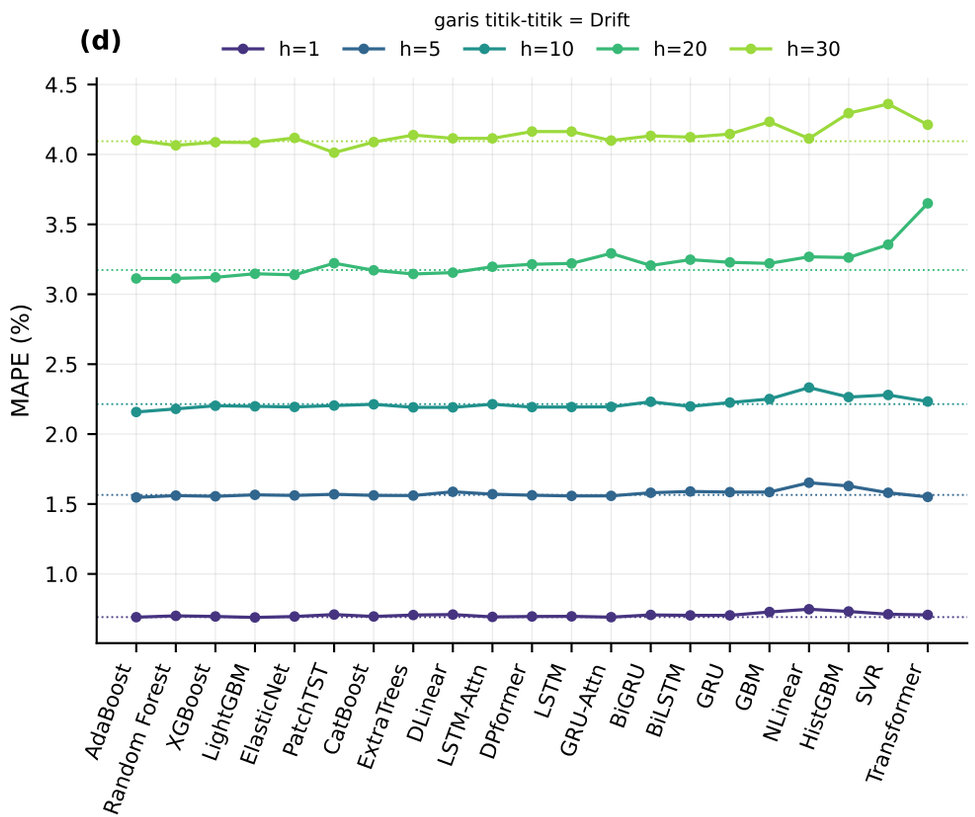

(a)
h = 20 | Multivariat
Aktual
Drift
DPformer
GRU-Attn
AdaBoost
Harga Antam (juta Rp/gram)
3.0
2.5
2.0
1.5
1.0
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
2025-05
2025-09
2026-01
2026-05
2026-09


(b)
RMSE (ribu Rp)
AdaBoost 21.8 46.0 64.4 91.3 120.7
Random Forest 21.9 46.9 66.8 94.5 124.2
ElasticNet 21.8 46.8 66.7 95.4 126.8
XGBoost 21.7 46.5 67.3 95.8 127.0
PatchTST 22.0 47.0 67.2 98.7 123.7
ExtraTrees 22.2 47.0 67.3 95.1 126.2
LightGBM 21.4 46.7 66.6 96.8 127.1
DLinear 22.1 47.6 67.3 96.9 126.4
LSTM-Attn 21.8 46.9 67.6 97.2 126.5
GRU-Attn 21.8 46.6 67.0 99.4 126.1
CatBoost 21.7 46.9 67.6 97.7 127.0
DPformer 21.8 46.7 66.9 97.7 128.6
LSTM 21.9 46.6 67.0 98.6 129.0
GRU 22.0 47.5 66.7 98.5 127.7
BiGRU 22.1 47.2 67.8 98.3 128.8
BiLSTM 22.1 47.4 67.2 98.9 127.9
GBM 22.7 48.2 68.3 96.7 127.2
NLinear 23.3 49.1 70.2 98.0 124.1
HistGBM 22.4 49.0 68.3 97.3 128.4
Transformer 22.1 45.5 67.6 111.1 129.6
SVR 22.1 46.8 69.4 103.5 135.1
h=1
h=5
h=10
h=20
h=30
RMSE vs Drift (%) [merah = lebih buruk]
10
5
0
−5
−10


(c)
AdaBoost
Random Forest
ElasticNet
XGBoost
PatchTST
ExtraTrees
LightGBM
DLinear
LSTM-Attn
GRU-Attn
CatBoost
DPformer
LSTM
GRU
BiGRU
BiLSTM
GBM
NLinear
HistGBM
Transformer
SVR
0
50000
100000
150000
200000
250000
MAE (Rp), ditumpuk antar-horizon
h=1
h=5
h=10
h=20
h=30


(d)
garis titik-titik = Drift
h=1
h=5
h=10
h=20
h=30
MAPE (%)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
AdaBoost
Random Forest
XGBoost
LightGBM
ElasticNet
PatchTST
CatBoost
ExtraTrees
DLinear
LSTM-Attn
DPformer
LSTM
GRU-Attn
BiGRU
BiLSTM
GRU
GBM
NLinear
HistGBM
SVR
Transformer